STAT 516: Homework 10

Instructions: Please submit your homework as a single PDF on Gradescope. After uploading, please make sure each page is assigned to the correct question before submitting. Show your work clearly.

1. **(10 points)** Let $X_1, X_2, \ldots, X_{36}$ and $Y_1, Y_2, \ldots, Y_{49}$ be independent random samples from distributions with means

   $$\mu_X = 30.4 \qquad \text{and} \qquad \mu_Y = 32.1,$$

   and standard deviations

   $$\sigma_X = 12 \qquad \text{and} \qquad \sigma_Y = 14.$$

   What is the approximate value of

   $$P\left(\overline{X}_{36} > \overline{Y}_{49}\right)?$$

2. **(10 points)** Let $X_1, \ldots, X_{100}$ be a random sample from an exponential distribution with mean $\frac{1}{2}$.

   Determine the approximate value of

   $$P\left(\sum_{i=1}^{100} X_i > 57\right).$$

   (Hint: using the Central Limit Theorem.)

3. **(10 points)** An insurance company sells 500 auto insurance policies. The number of claims per year for each policy follows a Poisson distribution with mean 0.2. Claim counts are independent for different policies.

   Using the normal approximation, calculate the probability of 80 or more claims.

4. **(10 points)** An insurance company issues 1250 vision care insurance policies. The number of claims filed by a policyholder under a vision care insurance policy during one year is a Poisson random variable with mean 2. Assume the numbers of claims filed by distinct policyholders are independent of one another.

   Calculate the approximate probability that there is a total of between 2450 and 2600 claims during a one-year period.

5. **(10 points)** The total claim amount for a health insurance policy follows a distribution with density function

   $$f(x) = \frac{1}{1000} e^{-x/1000}, \qquad x > 0.$$

   The premium for the policy is set at the expected total claim amount plus 100. If 100 policies are sold, calculate the approximate probability that the insurance company will have claims exceeding the premiums collected.

6. **(10 points)** Let X be a continuous random variable with density function

$$f(x) = \begin{cases} \dfrac{2}{x^2}, & 1 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

If

$$Y = X^{1/2},$$

what is the density function of Y for $1 < y < \sqrt{2}$?

7. **(10 points)** Let X be a continuous random variable with density function

$$f(x) = \begin{cases} 2e^{-2x}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Let

$$Y = e^{-X}.$$

What is the density function of Y?

8. **(10 points)** Let X be a continuous random variable with density function

$$f(x) = \begin{cases} \dfrac{1}{4}x^3, & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Let

$$Y = 8 - X^3.$$

Find the density function of Y.

9. **(10 points)** Let X be a continuous random variable with density function

$$f(x) = \begin{cases} 1 - |x|, & -1 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Let

$$Y = X^2.$$

Find the density function of Y.

10. **(10 points)** Let U be uniformly distributed on $[0, 1]$. Determine the density function of $Y = U^4$ for $0 \le Y \le 1$.